

MUHAMMAD ADITYA

Software Engineer

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PROFESSIONAL SUMMARY

Computer engineering graduate and technical lead with hands-on experience building and operating an integrated security technology platform end to end – from Node.js backend services and Flutter mobile applications to a self-managed Kubernetes cluster on Proxmox VE. Currently technical lead for JAGRA, a multi-service HR and security platform covering payroll, digital attendance, real-time guard patrol, and AI-based CCTV monitoring, and technical expert for drone-based computer vision inspection systems at RA ATAP Telkom University. Experienced in enterprise client integration, hardware evaluation (GPS telematics, body-worn cameras, CCTV), and technical documentation, with a continuing focus on network security and IT governance.

TECHNICAL SKILLS

Infrastructure & DevOps: Kubernetes, Proxmox VE, Linux, containerd, PM2, Prometheus, Grafana, Cloudflare Tunnel, MinIO, Redis, SSH key management, subnetting & network segmentation

Backend & Web: Node.js, Express.js, Next.js, REST API design, Firebase Cloud Messaging (FCM), MQTT

Mobile: Flutter, Dart, Android SDK, Google Play Console release management (AAB, permissions, Data Safety)

Security & Networking: Network security, MikroTik RouterOS, Kali Linux, vulnerability identification, access control systems

AI & Computer Vision: Roboflow (dataset annotation, training, deployment), image processing, real-time object detection, model inference on edge/drone environments, Google Colab

Hardware & IoT: UAV / drone systems, Dahua MPT220 body-worn cameras & EEC300 docking stations, Arduino, Proteus, Autodesk Eagle, piezoelectric & geophone sensors

Programming: Python, JavaScript, Dart, C++, HTML, CSS etc.

Tools: Git & GitHub, VS Code, GitHub Copilot, Claude Code, 9Router etc.

NOW - VIBECODE

PROFESSIONAL EXPERIENCE

Technical Lead – JAGRA Integrated Security Platform

PT Global Andalan Solusi / JAGRA | [Jan 2026] – Present

- Lead technical development, DevOps, and documentation for JAGRA, a multi-service platform covering HR management (payroll, digital attendance), real-time guard patrol, and AI-based CCTV monitoring.
- Build and maintain Node.js/Express backend services on a self-managed Kubernetes cluster hosted on Proxmox VE, supported by Redis, MinIO object storage, a Next.js front end, and Cloudflare Tunnel ingress.
- Deployed a full observability stack (Prometheus and Grafana via the kube-prometheus-stack Helm chart) for cluster and service monitoring, sustaining production uptime beyond 60 days with low resource utilisation.
- Expanded platform infrastructure by provisioning an additional HPE ProLiant node with Proxmox VE, including internal DNS, subnet planning, and SSH key-based access across all cluster nodes.
- Released the Jagra Patrol Flutter application to the Google Play Store, resolving foreground service and full-screen intent permission declarations, Data Safety disclosures, AAB signing, and version management.
- Delivered enterprise client integration for Kawasan Berikat Nusantara (KBN): live video streaming, SOS panic-button workflow, and Dahua MPT220 body-worn camera evidence management over a Wi-Fi upload path, including EEC300 docking station architecture and API documentation.
- Evaluated and specified GPS telematics hardware (Teltonika FMC920, Superspring) for patrol vehicle tracking, producing bill of materials, integration architecture, and client-facing technical documents.
- Manage a multi-repository Git architecture with submodules, PAT authentication, and branch policy for a distributed development team, and produce feature storyboards and product documentation for stakeholders.

Software Engineer – Drone & Computer Vision

RA ATAP, Telkom University – Bandung | Sep 2025 – Jan 2026

- Promoted from intern to technical expert on the strength of delivery performance during the internship period.
- Lead development of drone-based inspection systems powered by computer vision, from model design to field deployment.
- Mentor and direct interns across software engineering, computer vision, and drone engineering projects, maintaining work standards, project timelines, and overall team output quality.
- Drive R&D on a new seismic-sensing drone programme comprising two airframes, one using piezoelectric sensors and one using geophone sensors.

Software Engineer Intern – Computer Vision & Drone Systems

RA ATAP, Telkom University – Bandung | Sep 2024 – Sep 2025

- Developed an infrastructure crack-detection system using Roboflow as the primary platform for dataset annotation, model training, and deployment.
- Built a drone-based flood detection model using aerial image processing for environmental monitoring.
- Implemented the full AI pipeline – dataset preparation, training, validation, and inference – on a constrained drone runtime environment.
- Integrated computer vision models into drone systems to enable real-time detection in the field.
- Contributed to a fixed-wing mapping project covering aerial image processing and flight testing.

IT Support & Service Technician Intern

CV ADHI – Yogyakarta | Nov 2019 – Jan 2020

- Serviced and troubleshooted laptops and PCs, covering operating system installation, hardware repair, and software maintenance; handled printer repairs for both mechanical and basic software faults.
- Acted as technical customer service: received and analysed customer complaints, provided technical solutions, and coordinated the repair process end to end.
- Rotated on a two-week cycle between technician and customer service duties, building flexibility across both technical delivery and client-facing communication.

EDUCATION

Diploma in Computer Engineering

Telkom University, Bandung | Sept 2022 – June 2025 | GPA 3.43 / 4.00

TRAINING

- **MikroTik Competency Certificate** – Computer Networking (Hands-on MikroTik), HIPOTESA Lamongan, 2021
- **CCNAv7: Enterprise Networking, Security, and Automation** – Cisco Networking Academy, 2024
- **HCIA-Security V4.0** – Huawei ICT Strategy & Marketing Department, 2024
- **EPrT (English Proficiency Test)** – Score 523, valid 2025–2027
- **Huawei Enterprise / Talent programmes** – Digital Power; Overview of IoT Technologies; AI Basic (CRA Training Program), 2024
- **Cybersecurity & IT governance training** – SIEM Detection and Response; Identifying Cyber Security Vulnerabilities in IT Assets; Ethical Hacking Techniques; AI-Driven IT Risk Strategy & Continuous Exposure Management; COBIT 2019 for Sustainable IT Governance (PROXSIS, 2024–2025)
- **Network Security with CTI Group** – public lecture, CTI Group, 2024
- **Introduction to Containerization Using Podman** – ID-Networkers, 2025
- Etc.

ORGANISATIONAL ACTIVITIES

- **Community Service (Abdimas), Kelurahan Batununggal, Bandung – 2025:** delivered a web-based traffic and flood monitoring application using drone imagery
- **Yayasan Qurota A'yun – IT Division Member (2025–Jan 2026):** support foundation IT needs including administration systems, digital publication, activity documentation, and device maintenance; collaborate across divisions on digitalisation initiatives
- **Yayasan Qurota A'yun – Head of Sports Division (2023–2024):** led and coordinated all sporting activities, managed training schedules and internal competitions, and supervised division performance